CIS Benchmarks

CIS Microsoft Windows Workstation Benchmark Build Kit ReadMe

ReadMe

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Introduction

The purpose of this document is to describe the components of the CIS Microsoft Windows Build Kit and provide instruction on how to implement it within your environment. The Build Kit is designed to cover the majority of the benchmark settings. **These templates are to be modified in alignment with your organization's defined policies.**

Note: Prior to applying a Build Kit, verify that the most recent Microsoft Windows Administrative Templates have been downloaded directly from Microsoft and applied to the reference system.

WARNING: Reviewing the content within the corresponding Benchmark PDF is imperative for an overall successful application of the Build Kit, as there may be some settings that your organization needs to exempt itself from, due to unique operational requirements.

Applying the Build Kit to a system without proper testing and review may result in a negative impact within your environment.

It is acceptable if 100% of the benchmark is not applied, as it is the responsibility and decision of each organization to determine which settings are applicable to their unique needs.

Build Kit Contents and Application

Inside each Build Kit you will find folders containing several GPOs and custom templates for settings not covered by the standard Microsoft Windows Administrative Templates package.

Prior to applying a Build Kit, verify that the most recent Microsoft Windows Administrative Templates have been downloaded directly from Microsoft and applied to the system.

BITLOCKER = BitLocker settings
Next Generation = Next Generation Windows Security (NGWS) settings
COMP = Workstation computer settings
USER = User settings (all platform types)
SERVICES-L1= Services settings for Level 1
SERVICES-L2= Services settings for Level 2
L1 = Level 1 profile
L2 = Level 2 profile
<*Name*> Template – Custom GPO Template

Depending on which profile you decide to apply (L1 or L2) here are the GPOs you will want to use:

CIS Group Policy Object (GPO) Name → Use case ↓	COMP-L1	COMP-L2	USER-L1	USER-L2	BITLOCKER	SERVICES-L1	SERVICES-L2	Next Generation
Level 1 Workstation	X		X			X		
Level 2 Workstation	X	X	X	X		X	X	
BitLocker					X			
Next Generation Window Security (DC,MS,WS)								X

Note: It may be helpful to think of Level 1 as a foundation and the other profiles (Level 2) as "add-ons" to Level 1. For example, if you intend to harden a system to Level 2, then you need to apply ***both*** Level 1 and Level 2 GPOs in order to be L2 compliant. BitLocker and Next Generation Windows Security are **ONLY** to be applied if you will be using them within an organization.

Application for Stand-alone & Cloud Systems

Certain settings will need to be tailored to your organizational need before applying to a standalone system.

It is imperative appropriate tailoring is done before applying build kits to systems, they should not be applied out of the box. The stand-alone benchmark build kit can be managed and applied using a variety of methods, however 3 example methods are shown here. This document will only cover them at a high level with the LGPO usage being covered in more detail below. It is up to the organization's systems administrator to decide how to automate, and deploy with the tools they may uniquely have access to.

1. Import Group Policies into a test/sandbox Domain Controller, tailor as needed, then export using the backup function.
    a. Apply to production stand-alone systems using LGPO manually or through automation (more detail below on LGPO)
2. Apply chosen profiles to a Windows 10 reference machine using LGPO, tailor as needed using local group policy editor, then image and deploy as desired with your software of choice.
3. Apply chosen profiles to a Windows 10 reference machine using LGPO, tailor as needed using local group policy editor, then use LGPO to backup settings to later be deployed to production systems via LGPO.

The default Administrator and Guest accounts have been renamed within the Build kits so make sure to verify you have tailored these settings as needed before applying to a standalone system or cloud system.

WARNING: The necessary modifications of particular settings will be dependent upon the function of the standalone/cloud system. **Most modifications will be conducted in the User Rights section of the appropriate computer settings GPO and within the Remote Access based recommendations.** Make sure that you have a backup administrator account created as the built in is disabled with the GPO out of the box. As standalone users are unique to each system, locking a system down by user may result in issues related to local file sharing. Additionally, Remote Desktop settings may be affected as locking down that system may result in insufficient inbound connectivity from another system.

Note: Before you begin, please test the below steps on test system(s). If the test system is a VM, take a snapshot before so you have something to revert back to. If hardware-based, take a full image backup that will be ready to be restored quickly.

LGPO usage and examples

1) Download and prep the CIS Build Kit for the particular O/S. i.e. Win2008, Win2008 R2, Win2012, Win2012 R2, Win2016, Win2019, Win7, Win8.1, Win10.
    a. Copy the Build Kit to the standalone system.
    b. Important: Unzip the files to a shorter and much smaller directory structure. An example of this is below. **Note:** The biggest issue people have with the Build Kits is that the files can get corrupted in the unzip process due to the long filenames.

EXAMPLES:

`C:\CIS\Win10\Computer L1\{`*Actual GUID*`}`
`C:\CIS\Win10\User L1\{`*Actual GUID*`}`

e.g.
`C:\CIS\Win10\Computer L1\{1CF79D2C-7EF9-4B79-81DF-39E865439DB6}`
`C:\CIS\Win10\User L1\{44A16D69-5D67-453F-98FE-9880ADD6E7E3}`

2) Download LGPO and install LGPO
    a. Install `LGPO.zip` from this link:
    b. https://www.microsoft.com/en-us/download/details.aspx?id=55319
    c. Unzip it to a directory such as `C:\CIS\LGPO\`.
    d. Read the included manual **LGPO.pdf** this includes details on the full functionality, syntax and use cases of the tool.

3) Running LGPO as an example
    a. Open a Command Prompt as an Administrator.
    b. Change to the directory where LGPO.exe is unzipped and run the following:
        i. This will install the **Computer L1 Profile** settings:
        `LGPO.exe /g "C:\CIS\Win10\Computer L1\{GUID}" /v > LGPO-CL1.out 2> LGPO-CL1.err`
        Note: Make sure to use a lowercase **`/g`** in the above command

        ii. This will install the **User** settings:
        `LGPO.exe /g "C:\CIS\Win10\User L1\{GUID}" /v > LGPO-UL1.out 2> LGPO-UL1.err`
        iii. Lastly open the output and error files (e.g. LGPO-CL1.out and LGPO-CL1.err) in a notepad and ensure there aren't any errors and the setting applied properly.

4) Reboot
5) Login with a local Administrator account
6) Check that the CIS policies have been applied by doing the following:
    a. Run `GPRESULT /H:GPOReport.html` at an Administrative Command Prompt.
    b. Copy the `GPOReport.html` file to somewhere like `C:\CIS\`
    c. Change to the above directory and type `GPOReport.html` and press [Enter] to view the file.
    d. Analyze the results.

Support

CIS offers free support with our memberships. In the event that you have an issue with our Build Kits or any of the other products, we offer you a few options.

If you notice any typos, bugs or incorrect settings within a Build Kit, Benchmark or assessment content, please create a ticket in the Windows Benchmark community here: https://workbench.cisecurity.org/communities/2.

If you have any questions about why a given recommendation is contained within a Benchmark or why we decided on checking for a specific setting please feel free to post this question directly within the discussions of the community (https://workbench.cisecurity.org/communities/2) so that one of the subject matter experts can answer the question.

If you need support getting the benchmarks/assessment/remediation working in your environment you can contact our Support Department at https://www.cisecurity.org/support/. A support representative will reach out quickly to help resolve your issue. Please make sure to provide as much information as possible to help expedite your support.